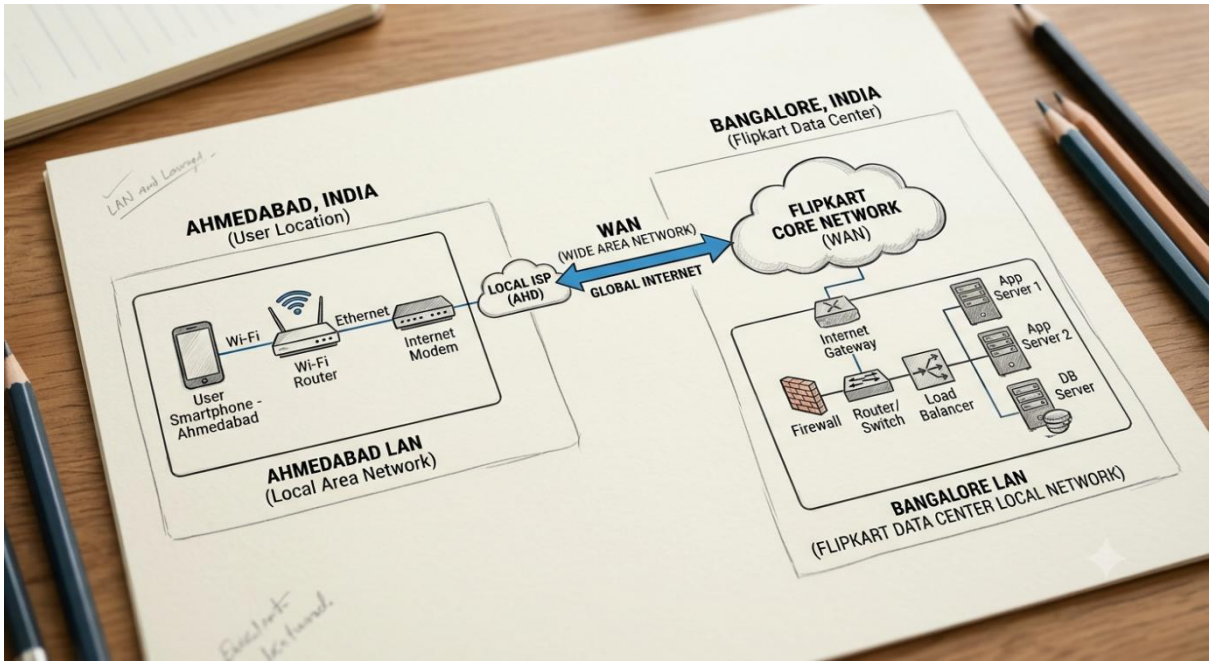
Day – 1 Introduction to WAN technologies theory.

Task – 1 Differences WAN and LAN.

- Key Differences Between LAN and WAN
- Geographic coverage: A LAN (Local Area Network) is restricted to a small, contained physical space like your home, a single office building, or a college campus. In contrast, a WAN (Wide Area Network) spans massive distances, connecting computers across different cities, countries, or even continents.
- Ownership and control: A LAN is typically private, set up and managed entirely by a single person, household, or organization (like your home Wi-Fi router). A WAN, however, relies on public infrastructure, telecommunication companies, and multiple Internet Service Providers (ISPs) working together to route traffic globally.
- Speed and bandwidth: Because data travels very short distances over dedicated cables or local wireless signals, a LAN offers exceptionally high speeds and ultra-low latency. A WAN usually experiences lower data transfer rates and higher latency because information has to hop across countless routers and long-distance lines.
- Real-World Examples
- LAN Example: Connecting your phone to your home Wi-Fi to instantly cast a YouTube video to your smart TV sitting in the living room, or printing a document from your laptop to a printer plugged into the same office network.
- WAN Example: Opening the Zomato app to order dinner or booking a train ticket on the IRCTC app. Your phone communicates via mobile data or broadband across the public internet (a WAN) to reach remote cloud servers located miles away in another city.

Task – 2 How to connect the router of far distance.



Task – 3 Explaining the MPLS .

- MPLS (Multiprotocol Label Switching) is a data-carrying mechanism designed to speed up and manage network traffic flow. Unlike traditional IP routing, where every router independently analyzes a packet's destination IP address and consults a complex routing table to determine the next hop, MPLS uses labels. When a packet enters an MPLS network, it is assigned a specific short label, and all subsequent routers route the packet based solely on this label, without needing to perform time-consuming IP lookups. This approach, known as label switching, significantly reduces the computational overhead on routers, leading to faster data forwarding and lower latency. Furthermore, MPLS allows for advanced traffic engineering, enabling network administrators to pre-define specific, optimized paths for different types of data, improving overall network efficiency and reliability compared to the simple "best-effort" nature of traditional IP routing.

Task – 4 Pre-config MPLS Network.

- Forwarding Based on Fixed Labels (LSR Lookups) Instead of Full IP Headers:

- In Standard IP Routing: Every router along the path inspects the packet's Layer 3 destination IP address, searches its routing table for the longest prefix match, and decides the next hop individually.
- In MPLS: The ingress router (Provider Edge / PE) attaches a short, fixed-length numerical label to the packet header. Intermediate routers (Label Switching Routers / LSRs) read only this label from a specialized Label Forwarding Information Base (LFIB), swap it with a new label, and forward it immediately without re-examining the IP destination header at every hop.
- □ Pre-determined Paths (LSPs) & Penultimate Hop Popping (PHP):
- Explicit Pathing / LSPs: Traffic follows a pre-established Label Switched Path (LSP) through the network core rather than dynamically calculating a hop-by-hop path based on standard IP metrics at every step.
- PHP Behavior: The second-to-last router before the destination (the penultimate hop) strips off the top MPLS label (using explicit/implicit null label instructions) before forwarding the raw packet to the final egress router. This removes the double-lookup burden on the egress router, allowing it to process the underlying IP packet directly.

Task – 5 Specific benefits MPLS.

- Key benefits of MPLS for high-traffic apps
- Guaranteed Quality of Service (QoS) and traffic prioritization: MPLS lets network operators tag and prioritize specific data streams. For a video streaming platform like YouTube, video packets can be given higher priority over background downloads or regular web browsing, ensuring smooth, buffer-free playback even when network traffic spikes during peak evening hours.
- Predictable low latency through label switching: Because MPLS routers forward packets using fixed short labels instead of executing complex IP header lookups at every single hop, it significantly reduces processing overhead and delay. For a platform like BookMyShow during a massive flash sale or concert ticket drop, this speed ensures faster transaction processing and prevents user timeouts.
- Reliable path control and traffic engineering: MPLS allows administrators to pre-engineer exact routes across the network core (Label Switched Paths). If a standard link gets congested, critical business traffic is dynamically routed along a pre-planned, optimized path, preventing packet loss, jitter, and unexpected downtime during high-demand surges.